

6 (a) $E = 0$ or $E_A = (-)E_B$ (at $x = 11$ cm)

B1

$$Q_A/x^2 = Q_B/(20-x)^2 = 11^2/9^2$$

C1

$$Q_A/Q_B \text{ or ratio} = 1.5$$

A1 [3]

or

$$E \propto Q \text{ because } r \text{ same or } E = Q/4\pi\epsilon_0 r^2 \text{ and } r \text{ same}$$

(B1)

$$Q_A/Q_B = 48/32$$

(C1)

$$Q_A/Q_B \text{ or ratio} = 1.5$$

(A1)

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(b) (i) for max. speed, $\Delta V = (0.76 - 0.18) \text{ V}$ or $\Delta V = 0.58 \text{ V}$

C1

$$q\Delta V = \frac{1}{2}mv^2$$

$$2 \times (1.60 \times 10^{-19}) \times 0.58 = \frac{1}{2} \times 4 \times 1.66 \times 10^{-27} \times v^2$$

C1

$$v^2 = 5.59 \times 10^7$$

$$v = 7.5 \times 10^3 \text{ ms}^{-1}$$

A1 [3]

(ii) $\Delta V = 0.22 \text{ V}$

C1

$$2 \times (1.60 \times 10^{-19}) \times 0.22 = \frac{1}{2} \times 4 \times 1.66 \times 10^{-27} \times v^2$$

$$v^2 = 2.12 \times 10^7$$

$$v = 4.6 \times 10^3 \text{ ms}^{-1}$$

A1 [2]